Lab 7: Fuzzing for Input Validation Bugs

Tool: Hypothesis (property-based fuzzing) with pytest. Module under test: student_grades.py (average_marks, parse_srn, short_name).

Repository: https://github.com/codenoww/L6_L7_16_student_grades/tree/main/Lab%207

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Step 2: Running the buggy application

Screenshot 1a: normal inputs work; average_marks crashes on "56,,57" (ValueError) and parse_srn crashes on "abc" (ValueError).

```
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades> cd "Lab 7"
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> python student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 78,85,92
Average Marks: 85.0

==== Running parse_srn ====
Enter an SRN (e.g. PES1UG23CS123): PES1UG23CS123
Campus, Year and Dept: ('PES1', 23, 'CS')

==== Running short_name ====
Enter first and last name (e.g. ravi kumar): ravi kumar
Short Name: R. Kumar
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> python student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 56,,57
Traceback (most recent call last):
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 29, in <module>
    main()
    ~~~~^^
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 15, in main
    result = average_marks(user_input)
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 2, in average_marks
    marks = [int(m) for m in marks_csv.split(',')]
             ~~~^^^
ValueError: invalid literal for int() with base 10: ''
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> python student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 78,85,92
Average Marks: 85.0

==== Running parse_srn ====
Enter an SRN (e.g. PES1UG23CS123): abc
Traceback (most recent call last):
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 29, in <module>
    main()
    ~~~~^^
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 20, in main
    result = parse_srn(user_input)
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 6, in parse_srn
    return srn[:4], int(srn[6:8]), srn[8:10]
```

Screenshot 1b: short_name crashes on "ravi" (ValueError: not enough values to unpack).

```
Traceback (most recent call last):
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 29, in <module>
    main()
    ~~~~^^
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 20, in main
    result = parse_srn(user_input)
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 6, in parse_srn
    return srn[:4], int(srn[6:8]), srn[8:10]
                    ~~~^^^^^^^^^^
ValueError: invalid literal for int() with base 10: ''
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> python student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 78,85,92
Average Marks: 85.0

==== Running parse_srn ====
Enter an SRN (e.g. PES1UG23CS123): PES1UG23CS123
Campus, Year and Dept: ('PES1', 23, 'CS')

==== Running short_name ====
Enter first and last name (e.g. ravi kumar): ravi
Traceback (most recent call last):
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 29, in <module>
    main()
    ~~~~^^
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 25, in main
    result = short_name(user_input)
  File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 9, in short_name
    first, last = full_name.split()
    ^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7>
```

Step 4: Hypothesis tests failing on the buggy code

Hypothesis found two distinct failing inputs for each function: the empty string and None. Result: 3 failed, 1 passed.

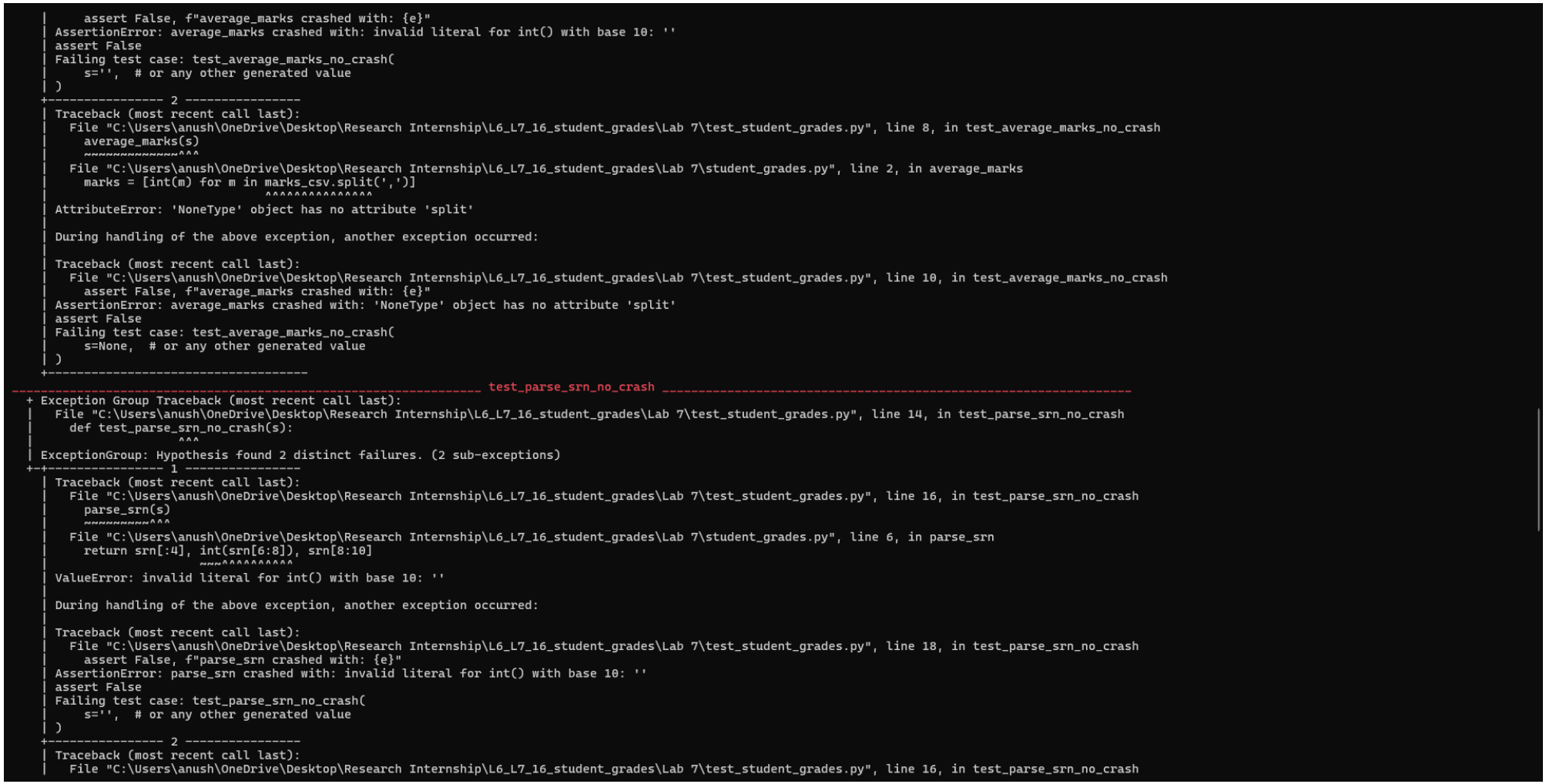
Screenshot 2a: test_average_marks_no_crash fails (s='' and s=None).

```
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> code .
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> python test_student_grades.py
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> python -m pytest test_student_grades.py
============================= test session starts =============================
platform win32 -- Python 3.14.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7
plugins: anyio-4.13.0, hypothesis-6.170.15
collected 4 items

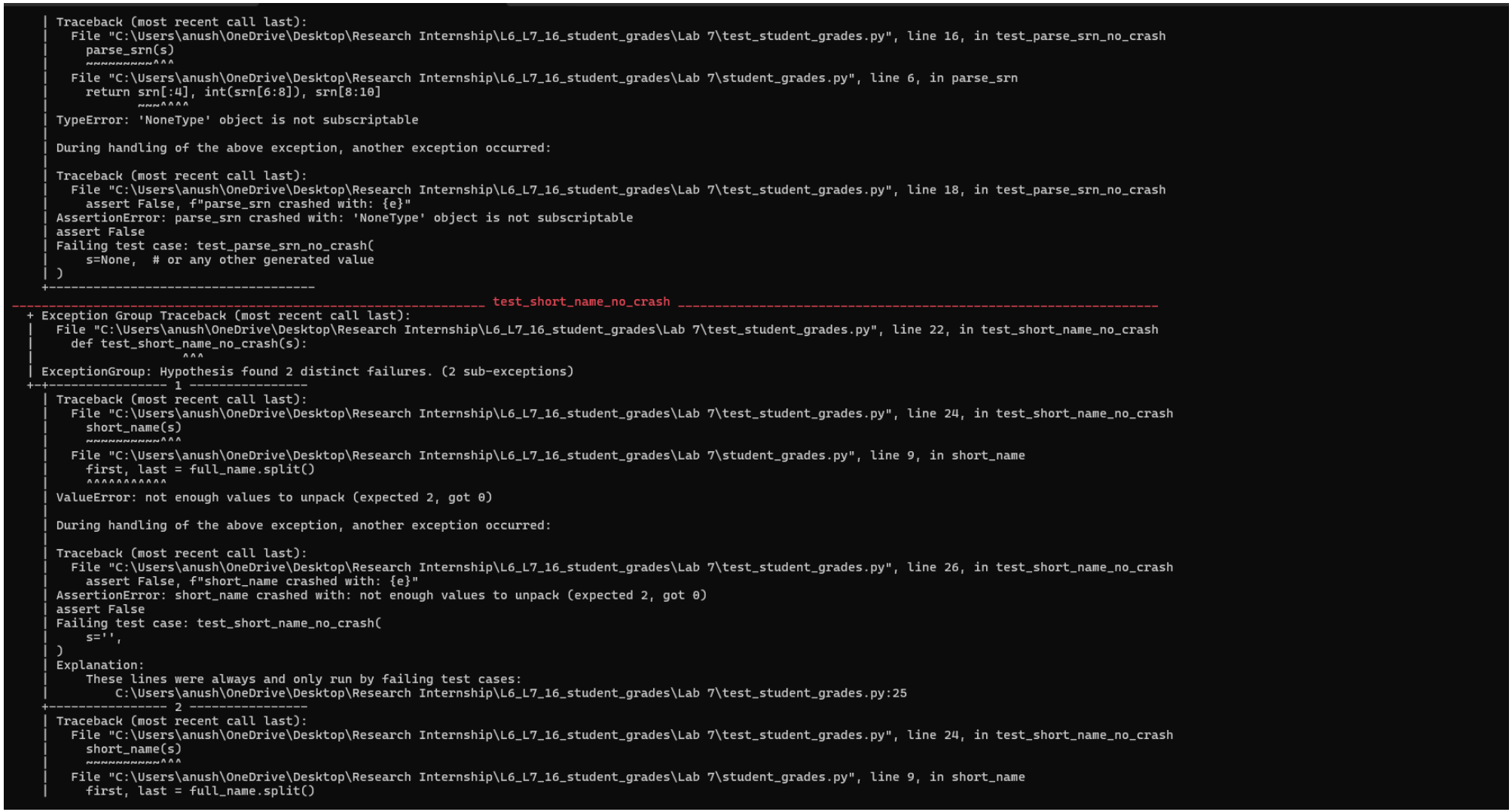
test_student_grades.py FFF.                                              [100%]

================================== FAILURES ===================================
_____________________ test_average_marks_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\test_student_grades.py", line 6, in test_average_marks_no_crash
  |     def test_average_marks_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\test_student_grades.py", line 8, in test_average_marks_no_crash
    |     average_marks(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 2, in average_marks
    |     marks = [int(m) for m in marks_csv.split(',')]
    |              ~~~^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\test_student_grades.py", line 10, in test_average_marks_no_crash
    |     assert False, f"average_marks crashed with: {e}"
    | AssertionError: average_marks crashed with: invalid literal for int() with base 10: ''
    | assert False
    | Failing test case: test_average_marks_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\test_student_grades.py", line 8, in test_average_marks_no_crash
    |     average_marks(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 2, in average_marks
    |     marks = [int(m) for m in marks_csv.split(',')]
    |                            ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\test_student_grades.py", line 10, in test_average_marks_no_crash
    |     assert False, f"average_marks crashed with: {e}"
    | AssertionError: average_marks crashed with: 'NoneType' object has no attribute 'split'
    | assert False
```

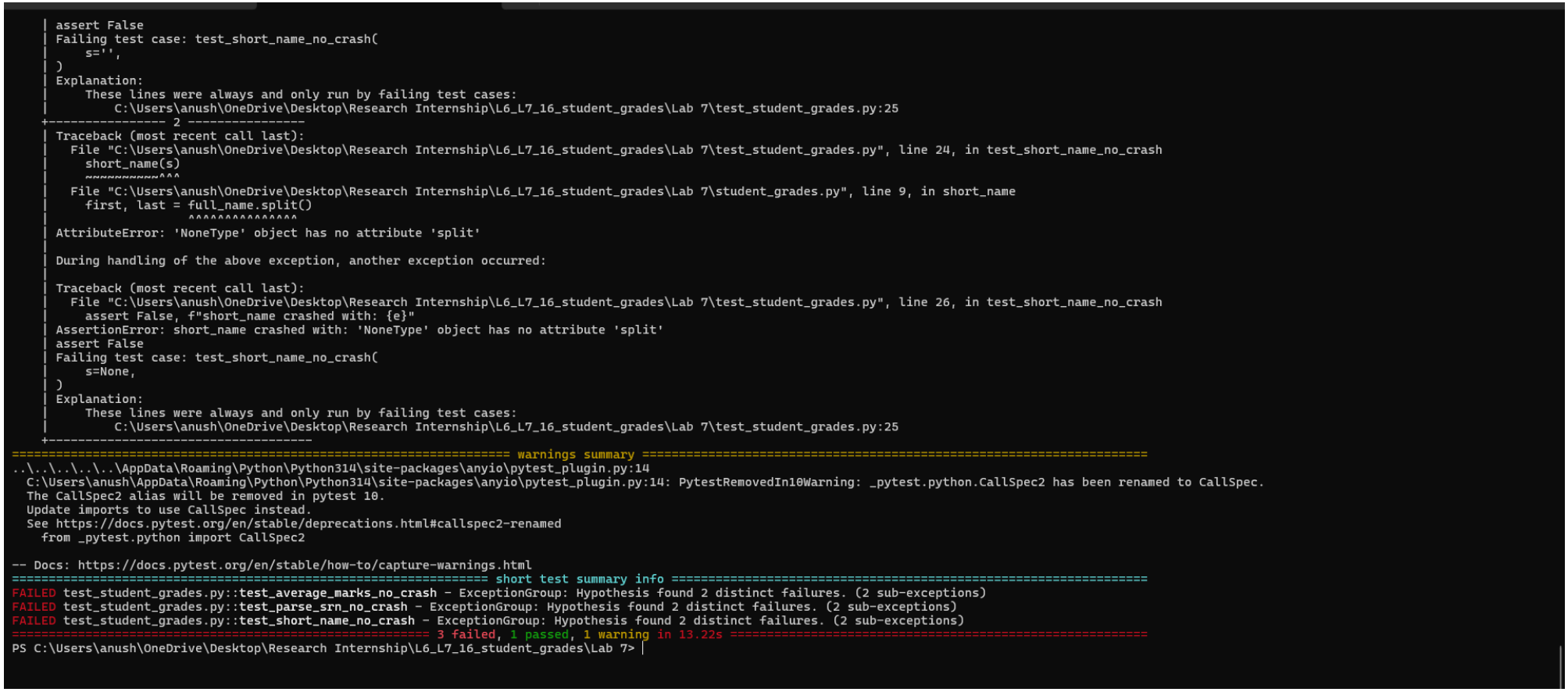
Screenshot 2b: average_marks failures continued; test_parse_srn_no_crash fails.



Screenshot 2c: parse_srn (None) and test_short_name_no_crash failures.



Screenshot 2d: failure summary: 3 failed, 1 passed.



```
    | assert False
    | Failing test case: test_short_name_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\test_student_grades.py:25
    +--------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\test_student_grades.py", line 24, in test_short_name_no_crash
    |     short_name(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\student_grades.py", line 9, in short_name
    |     first, last = full_name.split()
    |                   ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\test_student_grades.py", line 26, in test_short_name_no_crash
    |     assert False, f"short_name crashed with: {e}"
    | AssertionError: short_name crashed with: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_short_name_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7\test_student_grades.py:25
    +-----------------------------------
============================================================ warnings summary =============================================================
..\..\..\..\..\AppData\Roaming\Python\Python314\site-packages\anyio\pytest_plugin.py:14
  C:\Users\anush\AppData\Roaming\Python\Python314\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========================================================== short test summary info ==========================================================
FAILED test_student_grades.py::test_average_marks_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_student_grades.py::test_parse_srn_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_student_grades.py::test_short_name_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
================================================== 3 failed, 1 passed, 1 warning in 13.22s ==================================================
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7>
```

Step 5: Fixed application

Each function now validates its input type and format and returns None for invalid input; main() prints "Invalid input" instead of crashing.

Screenshot 3: fixed student_grades.py handles the same bad inputs without crashing.

```
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> code .
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> python student_grades.py
==== Running average_marks ====
Enter marks as CSV (e.g. 78,85,92): 56,,76
Average Marks: Invalid input

==== Running parse_srn ====
Enter an SRN (e.g. PES1UG23CS123): abc
Campus, Year and Dept: Invalid input

==== Running short_name ====
Enter first and last name (e.g. ravi kumar): ravi
Short Name: Invalid input
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7>
```

Step 6: Hypothesis tests on the fixed code

Screenshot 4: all 4 tests pass (4 passed).

```
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7> python -m pytest test_student_grades.py
============================= test session starts =============================
platform win32 -- Python 3.14.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7
plugins: anyio-4.13.0, hypothesis-6.170.15
collected 4 items

test_student_grades.py ....                                              [100%]

============================== warnings summary ===============================
..\..\..\..\..\AppData\Roaming\Python\Python314\site-packages\anyio\pytest_plugin.py:14
  C:\Users\anush\AppData\Roaming\Python\Python314\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
======================== 4 passed, 1 warning in 5.68s =========================
PS C:\Users\anush\OneDrive\Desktop\Research Internship\L6_L7_16_student_grades\Lab 7>
```